

Theory of the Diffraction of Electrons by Crystals

H. Bethe

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§ 1. The Refractive Index for de Broglie Waves

In recent times the wave nature of the electron has been directly demonstrated many times through beautiful experimental investigations. In all of these experiments we are dealing with the *crystal diffraction* of electrons either by *single crystals* [Davisson and Germer 1)] or by *microcrystalline* metals [Thomson 2), Rupp 3)]. Here we shall be concerned chiefly with the theoretical interpretation of the first-mentioned works, but the results are also in part applicable to diffraction by irregularly ordered microcrystals.

In the experiments of Davisson and Germer an electron beam of homogeneous velocity v falls perpendicularly upon the (111) face (octahedral face) of a nickel single crystal, and the electrons reflected back toward the entrance side are observed. It turns out that reflection occurs *selectively*, just as with X-rays: For a given de Broglie wavelength $\Lambda = \frac{h}{mv}$ (in other words, given velocity v) of the incident beam, one generally observes only continuous scattering in *all* directions; only at certain *discrete* incident wavelengths are the electrons preferentially diffracted into one *particular* direction 4) or into several symmetry-related directions.

Footnotes:

- 1) C. J. Davisson and Germer, Nature 119. p. 558. 1927; Phys. Review 30. p. 705. 1927; Proc. Nat. Ac. Amer. 14. p. 317. 1928. (D. and G. I, II, III.)
- 2) G. P. Thomson, Proc. Roy. Soc., A., 117. p. 600. 1928.
- 3) E. Rupp, Ann. d. Phys. 85. p. 981. 1928.
- 4) We specify the direction of the reflected beam by the angle Θ between it and the crystal normal, and by an azimuthal angle φ . Following Davisson and Germer, we denote the latter by the Miller indices of the simplest crystal face whose normal lies in the plane of reflection determined by the incident and reflected beams.
For longer electron waves, selective reflection comes into consideration chiefly in the (100) azimuth ($\varphi = 0, 120, 240^\circ$), in the (111) azimuth ($\varphi = 60, 180, 300^\circ$) and in the (110) azimuth ($\varphi = 30, 90, 150, 210, 270, 330^\circ$).

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From this phenomenon of selective reflection one may conclude that the electron waves are diffracted by the crystal lattice of nickel.

Qualitatively, then, there is complete analogy between the reflection of electrons and that of X-rays; quantitatively, however, the analogy is only partial: The relation between the *direction* of the diffracted beams and the de Broglie wavelength in vacuum, $\Lambda = \frac{h}{mv}$, remains unchanged

The other possible assumption is that the wavelength of the de Broglie wave, which outside the crystal is Λ , has a different value, λ , inside it, i.e. that the crystal must be assigned a refractive index

$$\mu = \frac{\Lambda}{\lambda} \quad (2)$$

for de Broglie waves. Such a refractive index was first introduced on purely phenomenological grounds by Eckart 2). The author 3) has independently of this

Footnotes:

- 1) A paper by Patterson (Nature 120. p. 46. 1927) deals theoretically with the possibility of a contraction factor.
- 2) C. Eckart, Proc. Nat. Acad. Washington 12. p. 460. 1927.
- 3) H. Bethe, Naturwissenschaften 15. p. 787. 1927 (cited as "Bethe I").

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shown that a refractive index is directly demanded by wave mechanics, and that the assumption of such an index is indeed in exact agreement with the experimental results: Relation (1) between vacuum wavelength Λ and diffraction angle Θ is preserved, because it already follows from the surface structure of the crystal alone; the magnitude of the selectively reflectable wavelengths, on the other hand, which depends on the three-dimensional action of the atoms in the interior of the *crystal*, is of course influenced by a refractive index attributed to the interior of the crystal. 1) For now the wavelength λ in the crystal must naturally have the magnitude following from lattice-diffraction theory and no longer the vacuum wavelength Λ . Since the vacuum wavelengths Λ follow from the measured electron velocity, and the crystal wavelengths λ can be computed from the known lattice of nickel, μ is determined as soon as it is decided which λ corresponds to a given Λ . Here again two possibilities arise: either one may assign to each wavelength that is selectively reflected according to the observations the next *larger* in the scale of the wavelengths λ that can be selectively reflected according to lattice theory: this assignment, indicated by dots in Fig. 1, was first proposed by Davisson and Germer 2) and yielded refractive indices $\mu < 1$. Against this, however, there are very grave objections, connected in large part with the wave-mechanical theory of the refractive index to be discussed presently 3), but also the circumstance that for $\mu < 1$ no electrons could emerge grazing from the crystal into the vacuum, whereas such "grazing beams" were observed by Davisson and Germer. 4) Consequently the experiments of Davisson and Germer admit only the possibility of relating the observed wavelengths Λ to the next *smaller* of the expected wavelengths λ .

Footnotes:

- 1) Cf. H. Bethe I, loc. cit.
- 2) The "new assignment" was presented by Davisson and Germer II, Phys. Rev. 30. p. 705 as possible but improbable. According to a letter of early June 1928, D. and G. have now likewise adopted the new assignment.
- 3) H. Bethe, Naturwissenschaften 16. p. 333. 1927. Cited as "Bethe II". [Tr.: misprint for "1928".]

each azimuth the longest selectively reflectable waves to be expected from lattice theory: In the (111) and (110) azimuths the longest waves of first order must suffer total reflection, and thus cannot be observed at all, whereas the corresponding electron beam in the (100) azimuth must emerge grazing into the vacuum. Precisely here, however, Davisson and Germer observed an especially strong beam. The agreement with theory is not quite so good in the higher orders.

Footnotes:

- 1) “Grazing beams” (grazingly emerging electron beams) were observed by Davisson and Germer in each of the three principal azimuths. Their occurrence is explained by Davisson and Germer in general by the fact that the path of the electrons from one atomic layer to the next is very large for grazing beam propagation, and that therefore a large part of the electrons scattered by the deeper layers of the crystal is absorbed before reaching the surface. Consequently the main part of the intensity of the observed beam is supplied by the electrons diffracted at the surface layer, and for the production of a grazingly emerging beam it is therefore not absolutely necessary that the waves scattered by all layers of the crystal be in phase with one another, i.e. that the Laue equations of space-lattice diffraction be satisfied. In this case, however, the intensity of the observed grazingly emerging beam will be especially large, and this is precisely the case in the (100) azimuth.

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Thus the longest second-order beam reflected by the (440) plane in the (111) azimuth should emerge grazing, but is not observed. This is probably due to the fact that in the second order the intensity falls off much more rapidly with decreasing Θ than in the first, and already at $\Theta = 70^\circ$, where reliable observations first become possible, the scattered waves of the uppermost atomic layer have exactly the opposite phase to those of the second. The absence of the beams to be expected at 232 and 248 volts in the (210) and (331) azimuths is also understandable because of their presumably small intensity. Another question is the irregularity of the third-order reflections in the (111) azimuth, which should agree with those in the (100) azimuth. The explanation for this follows only from the dynamical theory (§ 9). On the whole one may well say that the phenomenon of the Laue spots missing because of *total reflection* and the *intensity* of the “grazing beams” form a further support of the “new assignment”, but in no case speak for the old.

§ 2. The Field of the de Broglie Waves Inside the Crystal

The elementary theory developed in the preceding section is based on Laue's purely wave-kinematical theory of the refractive index.

Like Laue's theory, the theory developed here can represent only a first *approximation*. This is already evident from the fact that we have summarily assumed the potential of the electrons in the crystal to be *constant*, whereas consistently we must represent it in a spatial *Fourier series*.¹⁾ If this circumstance is taken into account, one is led to a *dynamical* theory corresponding to Ewald's dynamical theory of X-ray reflections²⁾, and largely following its method.

The course of the solution is roughly as follows: One represents the wave function as a superposition of plane waves, mathematically speaking: One expands ψ in a similar

Footnotes:

- 1) At the center of the atoms V even becomes infinite as $1/r$, and consequently the wavelength becomes infinitely small as $r^{1/2}$. The ansatz we use here contains this fact implicitly.
- 2) P. P. Ewald, Ann. d. Phys. 54. p. 514. 1917 (cited as Ewald III). [Tr.: misprint for "p. 519".]

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Fourier series to that of the potential, but with the wave number of the incident beam in the crystal, k_0 , still undetermined. Then between the amplitudes of the individual plane waves (the coefficients of the Fourier expansion of ψ) just as many equations must be satisfied as there are plane waves. Into these equations enter, besides the Fourier coefficients of the potential, which we must assume known, the known vacuum wave number K and the unknown wave number in the crystal (k_0) of the incident wave. In addition, doubly infinitely many boundary conditions must be satisfied. The solution of the indicated system of equations proceeds in such a way that the two amplitudes of the incident and the reflected wave are assumed *large* compared with the amplitudes of the remaining plane waves of the ψ -field, and all remaining amplitudes are expressed through these two. One then obtains two homogeneous equations between the two amplitudes assumed "large". The vanishing of their determinant yields, as usual, the dispersion equation, from which two different values result for the as yet unknown wave number k_0 . By a combination of two solutions with the two possible wave numbers k_0 , the boundary conditions can be satisfied, as in Ewald's theory. From the solution thus obtained one then determines the more accurate magnitude of the selectively reflectable wavelengths, the intensity of the reflected beam, and the resolving and reflecting power of the crystal.

As the origin of our coordinate system we choose the nucleus of any atom of the crystal surface. As coordinate axes a_1 , a_2 and a_3 we choose, say, the usual crystallographic axes, which moreover is not absolutely necessary. If we are dealing with a crystal of a system with orthogonal axes, we may of course assume a Cartesian coordinate system X , Y , Z . We shall also do this occasionally later, but where possible we choose the vectorial notation, which can be applied to any obliquely-angled axis system.

Footnotes:

- 1) The wave number (per unit length) is defined as 2π divided by wavelength.

Footnotes:

- 1) Ewald III, loc. cit., p. 536, and especially Ewald, Ztschr. f. Kristallographie 56. p. 129. 1921 (cited as “Ewald IV”).

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with real coefficients v_g . For crystals with orthogonal axes (rhombic, tetragonal, cubic, and, with suitable choice of axes, also hexagonal system) and identity distances a_1 , a_2 and a_3 , the reciprocal axes become

$$\mathbf{b}_i = \frac{\mathbf{a}_i}{a_i^2}, \quad b_i = \frac{1}{a_i} \quad (9b)$$

and the potential becomes simply:

$$U(x, y, z) = \sum_g v_g e^{2\pi i \left(\frac{g_1 x}{a_1} + \frac{g_2 y}{a_2} + \frac{g_3 z}{a_3} \right)} \quad (7c)$$

We now wish to consider the field of the de Broglie waves in the crystal. To do so we first assume the crystal to be infinitely extended, i.e. we regard the ansatz (7) for the potential as valid in all space. Later we shall then set up the conditions that are added by the existence of a surface of the crystal.

The most straightforward approach would be to solve the Schrödinger differential equation

$$\Delta\psi + \frac{8\pi^2 me}{h^2} (E + V)\psi = 0 \quad (3)$$

by the ansatz of a plane wave

$$\psi_0 = c_0 e^{i(\mathbf{f}_0 \mathbf{r})} \quad (10)$$

Here, following Ewald 1), we call $\mathbf{f}_0$ the propagation vector of the plane wave (10). It gives the direction of propagation of the plane wave; its absolute value k_0 is called the wave number and stands in the simple relation to the wavelength λ_0 of the plane wave

$$k_0 = \frac{2\pi}{\lambda_0} \quad (10a)$$

Substituting the ansatz (10) into the Schrödinger equation (3) and taking into account the Fourier expansion (7) of the potential, we w

$$\left(\frac{8\pi^2 me}{h^2} E - k_0^2 \right) c_0 e^{i(\mathbf{f}_0 \mathbf{r})} + \sum_g c_0 v_g e^{i(\mathbf{f}_0 \mathbf{r}) + 2\pi i(\mathbf{g} \mathbf{r})} = 0.$$

This equation cannot be satisfied simultaneously at all field points $\mathbf{r}$. Instead, it turns out

Since this equation holds for every field point $\mathbf{r}$, the coefficient of each exponential must vanish separately. This yields, with a corresponding change of the summation indices:

$$\psi_h(K^2 - k_h^2) + \sum_g v_g \psi_{h-g} = 0$$

or, if we extract from the sum the term $v_0 \psi_h$:

$$\psi_h(K^2 + v_0 - k_h^2) + \sum'_g v_g \psi_{h-g} = 0, \quad (13)$$

where the prime denotes that v_{000} is to be excluded.

- (13) holds for every index triple h_1, h_2, h_3 and thus represents a triply infinite scheme of equations valid inside the crystal, in each equation the amplitude of a ψ -oscillation ψ_h occurring multiplied by a resonance factor, all other ψ -amplitudes multiplied each by a Fourier coefficient of the potential. The system of equations represents the *exact* solution of the Schrödinger equation inside crystals, only inelastic collisions being disregarded. This distinguishes our method fundamentally from Born's 1), which seeks to grasp the processes in the collision of an electron with an atom by successive approximation. That Born's method is not adapted to the treatment of crystal diffraction is shown by the fact that Zwicky 2) could explain from the first approximation according to Born's method only the occurrence of crystal diffraction at all, but *not* the refractive index for the de Broglie wave. On the other hand, the method developed here is *not* applicable to diffraction by a *single atom* 3), but is essentially conditioned by the regular *arrangement* of the atoms in the crystal, as a consequence of which *diffraction by a single atom* whose structure is so far not sufficiently known, *recedes* behind diffraction by the *arrangement* of the atoms.

Our system of equations (13) is the necessary condition for the existence of “dynamical equilibrium” in the crystal, i.e. for the wave field ψ being continually regenerated by diffraction at the crystal atoms. The only remaining freedom is

Footnotes:

- 1) M. Born, Ztschr. f. Phys. 38. p. 803. 1926.
- 2) F. Zwicky, Proc. Nat. Acad. Amer. 12. p. 518. 1927. [Tr.: sic; correct is Sci. 13, pp. 518-525.]
- 3) Closest to our method comes that developed by Faxén and Holtsmark for diffraction by a single atom, which likewise refers to the actual ψ -field of the diffracted electron in the diffracting atom. Ztschr. f. Phys. 45. p. 307. 1927.

the choice of the propagation vector $\mathbf{f}_0$ and the amplitude ψ_0 of the primary wave; but only so long as the crystal is really infinitely extended and nowhere comes into contact with another medium. The surface phenomena treated in the following section determine the constants of the “primary beam”.

of the electron beam is directly proportional to the cosine of the angle between it and the normal to the crystal surface (cf. Fig. 4), the

[Figure 4: diagram of two beam bundles (incident and diffracted) meeting a horizontal crystal surface, with the surface normal drawn dashed and the angles ϑ_1 , ϑ_2 between the beams and the normal indicated.] Cross section of the beam bundles and direction cosines: $\frac{q_1}{q_2} = \frac{\cos \vartheta_1}{\cos \vartheta_2}$. Fig. 4

product of both quantities is therefore proportional to the normal component of the propagation vector. In this wave description, the electron number density is proportional to the squared modulus of the de Broglie-wave amplitude. We denote the particle flux of the primary beam in vacuum

$$C_0^2 \Gamma_1 = S_0 \quad (56)$$

and now have to distinguish *two cases*: 1. Outside the region of selective reflection¹⁾ we have $W > 1$, hence η and ω real. The intensities of the primary beam

$$s_1 = \frac{W^2 - \cos^2 \omega(H - z)}{W^2 - \cos^2 \omega H} \cdot S_0 \quad (56a)$$

Footnotes: 1) The same considerations always hold for the Laue case (cf. p. 79).

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and of the secondary beam

$$s_2 = \frac{\sin^2 \omega(H - z)}{W^2 - \cos^2 \omega H} \cdot S_0 \quad (56b)$$

are then subject to periodic fluctuations in the crystal, in such a way that in the end the number of electrons transported through a cross section of the crystal in the direction of the primary beam,

$$s = s_1 - s_2 = \frac{W^2 - 1}{W^2 - \cos^2 \omega H} \cdot S_0 \quad (56c)$$

is independent of the particular choice of the cross section. This is of course already required because otherwise in certain layers of the crystal there would be a permanent inflow, in others a permanent loss of electrons, whereas the entire electron wave field we are considering is supposed to be *stationary*. Yet the currents of the primary and secondary beams individually increase and decrease spatially periodically, both in the same sense, the period of such an intensity beat,

$$\frac{\pi}{\omega} = \frac{2\pi \cos \vartheta_2}{w\sqrt{W^2 - 1}}$$

amounting to a few atomic layers. This finding is analogous to Ewald's *Pendellösung*¹⁾ in the case of real resonance errors. In connection with the periodic intensity fluctuations of the electron beams in the crystal, the intensity of the observable secondary beam reflected back into the vacuum,

$$S_2 = S_0 \frac{\sin^2 \omega H}{W^2 - \cos^2 \omega H} \quad (57)$$

quantities of both processes, there nevertheless exist far-reaching differences. The reason for this is actually only a quantitative one: The influence of the crystal fields on the electron waves is, at equal wavelength, much greater than that of the “oscillating dipoles” of the crystal on the X-rays. This already shows in the refractive index, which deviates from unity by some hundredths for electrons, by only millionths for X-rays. Just as v_0 is not negligible compared to $\mathfrak{K}^2$ in the expression for the refractive index

$$\mu = \sqrt{1 + \frac{V_0}{E}} = \sqrt{1 + \frac{v_0}{\mathfrak{K}^2}}$$

so the remaining Fourier coefficients $v_{\mathfrak{g}}$ also play a role compared with the resonance denominators $\mathfrak{K}^2 - k_{\mathfrak{g}}^2$ of the “weak” beams. This has the consequence that the de Broglie waves of the field in the crystal for which the resonance denominators are *not* small nevertheless still possess a not negligible intensity and can react back on the strong beams. This reaction is expressed in the “dynamical potential quantities” V_{11} , V_{12} , V_{22} . In the X-ray interference theory, on the other hand, the intensity of the waves $f_{\mathfrak{g}}$ decreases so rapidly with the distance of the corresponding reciprocal lattice point from the sphere of propagation that the intensity of the “weak beams”

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becomes vanishingly small, no “reaction” need be taken into account. The same occurs for electrons of much smaller wavelengths, i.e. about in the region investigated by G. P. Thomson of $E = 15$ to 60 kV corresponding to $\lambda = 0.1$ to 0.05 Å.E. The sums in the potential quantities disappear, and in the case of a reflected beam $V_{11} = V_{22} = 0$ and $V_{12} = -v_{\mathfrak{h}}$.

The “reaction” of the weak beams on primary and secondary beam has first of all the consequence that the magnitude of the selectively reflectable wavelength is determined not alone by the refractive index, but in addition also by all Fourier coefficients of the potential $v_{\mathfrak{g}}$, i.e. by the whole potential variation. For X-rays and for electrons of very high velocity these quantities are already determined by the refractive index.

Loosely connected with the reaction is the small resolving power and large reflecting power of the crystal for de Broglie waves.

Furthermore, for electrons the expressions for the integrated reflecting power of a lattice plane $(h_1 h_2 h_3)$, i.e. for the reflected intensity on bombardment with electrons of a larger velocity range, become considerably more complicated. We found for this intensity in § 6

$$\int S_2 d\mathfrak{K} = \frac{\pi S_0 V_{12} \sqrt{\cos \vartheta}}{\mathfrak{K}(1 + \cos \vartheta)} = \frac{\pi S_0 \sqrt{\cos \vartheta}}{\mathfrak{K}(1 + \cos \vartheta)} \left(\sum_{\mathfrak{g}} \frac{v_{\mathfrak{g}} v_{\mathfrak{h}-\mathfrak{g}}}{\mathfrak{K}^2 - k_{\mathfrak{g}}^2} - v_{\mathfrak{h}} \right). \quad (62)$$

Here the “reaction” of the weak beams enters again. For X-rays, as mentioned above, the “integrated reflecting power” is simply proportional to $\varrho_{\mathfrak{h}}^E$. The same holds again for very fast electrons: V_{12} becomes there equal to $-v_{\mathfrak{h}}$.

The extinction laws too no longer agree under certain circumstances for electron and X-ray diffraction. The X-ray reflection $(h_1 h_2 h_3)$ is known to be extinguished when the corresponding point of the reciprocal lattice has weight zero (Ewald IV), i.e. when

$$\varrho_{\mathfrak{h}}^E = \varrho_{\mathfrak{h}}^K = v_{\mathfrak{h}} = 0$$

holds. Then of course the electron wave $\psi_{\mathfrak{h}}$

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can no longer couple directly to the primary wave. For in the dynamical equation

$$\psi_{\mathfrak{h}}(\kappa^2 - k_{\mathfrak{h}}^2) = - \sum \psi_{\mathfrak{g}} v_{\mathfrak{h}-\mathfrak{g}}$$

the amplitude ψ_0 of the primary wave is afflicted with the factor $v_{\mathfrak{h}}$. But it can happen under certain circumstances that in the sum on the right-hand side a non-vanishing amplitude $\psi_{\mathfrak{g}}$ occurs with a coefficient $v_{\mathfrak{h}-\mathfrak{g}}$ different from zero, and that thereby $\psi_{\mathfrak{h}}$ becomes indirectly of finite magnitude. The electron reflection ($h_1 h_2 h_3$) will then not be extinguished even for vanishing Fourier coefficient $v_{\mathfrak{h}}$, if two coefficients $v_{\mathfrak{g}}$ and $v_{\mathfrak{h}-\mathfrak{g}}$ are simultaneously different from zero.

For the very simplest structures such a breaking of the extinction laws cannot of course come into question. For the base-centered lattice, e.g., X-ray reflections are extinguished when $h_1 + h_2$ is odd. Thus, in order that $v_{\mathfrak{g}} \neq 0$, $g_1 + g_2 \equiv 0 \pmod{2}$ must hold, and in order that $v_{\mathfrak{h}-\mathfrak{g}} \neq 0$, $(h-g)_1 + (h-g)_2 \equiv 0 \pmod{2}$ must hold. From this, however, follows $h_1 + h_2 \equiv 0 \pmod{2}$, hence $v_{\mathfrak{h}} \neq 0$. Here therefore no electron reflection can occur through the “reaction” of weak beams when the corresponding X-ray reflection is extinguished. The same holds e.g. for the face-centered and for the body-centered lattice.

If, on the other hand, extinction is observed for X-rays when a definite index or an index sum (or difference) is divisible by a certain number p , then this extinction law is in general broken for electron diffraction. Let e.g. ($h_1 h_2 h_3$) be extinguished when

$$h_1 \equiv rp \pmod{np},$$

where p denotes an integer > 1 , and r can run through one or several numbers in the series $0, 1 \dots n-1$.¹⁾ Let, on the other hand,

$$h_1 \not\equiv 0 \pmod{p}$$

Footnotes: 1) The two other indices h_2 and h_3 may thereby still be subject to fairly arbitrary conditions which do not interest us here. Exactly the same holds of course when e.g. the extinction law reads $h_1 + h_2 \equiv rp \pmod{np}$, etc.

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— under the same conditions for h_2, h_3 — not extinguished. Then always

$$\begin{aligned} v_{\mathfrak{g}} &\neq 0, & \text{if } g_1 &\not\equiv 0 \pmod{r}, \\ v_{\mathfrak{h}-\mathfrak{g}} &\neq 0, & \text{if } (h-g)_1 &\not\equiv 0 \pmod{p}, \end{aligned}$$

i.e. for instance

$$(h-g)_1 \equiv rp - g_1 \pmod{np}.$$

The electron reflection $\mathfrak{h}$ thus occurs although $h_1 \equiv rp \pmod{np}$, i.e. $v_{\mathfrak{h}} = 0$.

An example of this is e.g. diamond. There are extinguished 1. all reflections with “mixed” (even and odd) indices, 2. all reflections for which $h_1 + h_2 + h_3 \equiv 2 \pmod{4}$.

The second law is broken for electrons, for e.g.

$$v_{\mathbf{g}} \neq 0, \quad \text{if} \quad g_1 + g_2 + g_3 \equiv 1 \pmod{4}$$

and

$$v_{\mathbf{h}-\mathbf{g}} \neq 0, \quad \text{if} \quad (h-g)_1 + (h-g)_2 + (h-g)_3 \equiv 1 \pmod{4}.$$

$\psi_{\mathbf{h}}$ thus occurs even when $h_1 + h_2 + h_3 \equiv 2 \pmod{4}$.¹⁾

SiC (carborundum II) behaves similarly. Here the X-ray diffraction $h_1 - h_2 \equiv 0 \pmod{3}$ is extinguished [unless $h_3 \equiv 0 \pmod{6}$, which is however irrelevant]. This extinction too no longer takes place in electron diffraction.

On the other hand all extinction laws

$$h_1 \not\equiv 0 \pmod{p}$$

and similar ones remain in force. For here $v_{\mathbf{g}}$ and $v_{\mathbf{h}-\mathbf{g}}$ are non-zero only when $g_1, (h-g)_1$ and consequently also h_1 are divisible by p . In this case electron interference beams can only arise when $v_{\mathbf{h}} \neq 0$, i.e. when they already arise directly from the primary beam.

For high electron velocities the “indirect reflection” due to weak-beam back-coupling

Footnotes: 1) It is not inconceivable that exactly the corresponding phenomenon is partly responsible for the imperfect extinction of the X-ray reflection (222) in diamond. Since diamond is an ideal crystal, the small influence of the “weak” beams e.g. (111), ($\bar{1}\bar{1}1$), etc., may suffice to produce the reflection (222) “indirectly”. *Translator’s notes on symbols: (i) The symbol printed as a swash italic z (e.g. in $\xi = 2rz \, na_0$ and in $(3Nz \, 8\pi)$) is transcribed exactly as printed; in context it plays the role of the effective nuclear charge / number of conduction electrons. (ii) In eq. (81) the subscripts of the last terms of G^2 and $\bar{g}$ are barely legible at this scan resolution (they look like “2”); they are transcribed as g_3 in agreement with the parallel definitions (80) and (80a) and the stated permutation symmetry of the three components. (iii) In the W_i equation the original prints z in $(3Nz \, 8\pi)$ but a capital $Z^{2/3}$ in the second half; the surrounding text denotes the number of conduction electrons per atom by z .*

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is of course no longer relevant. At low velocities it may be observed, provided a conducting crystal with a suitable structure is found. However, the inelastic collisions strongly attenuate all secondary beams $\psi_{\mathbf{g}}$, and this attenuation occurs twice for the “indirectly reflected” beams, since (in Bragg’s picture) they must be thought of as arising from twofold lattice-plane reflection of the primary beam. Their intensity will therefore in practice be considerably smaller than that of the directly reflected beams, although according to the theory developed here it should be of about the same intensity as these.

§ 9. Symmetric Reflection. Coordinate Transformation

I would like to express my deepest thanks to Prof. Sommerfeld for suggesting this work and for his continued interest and encouragement during its progress.

Summary

According to wave mechanics, the diffraction of

Footnotes: 1) W. Houston, Ztschr. f. Phys. 48. p. 449. 1928.

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electrons by crystals observed by Davisson and Germer, Rupp, Thomson, etc., is treated. In the first approximation, the observed discrepancy between X-ray and electron diffraction can be explained by a refractive index. From the observed refractive indices one may infer, for nickel, an accelerating potential $V_0 = 14.8$ volts which the crystal exerts on electrons (§ 1). The dynamics of the field of the de Broglie waves in the crystal (§ 2) and the conditions at its surface (§ 3) are developed. With the simplifying assumption that, besides the primary beam, only *one* strong diffracted beam occurs (§ 4), the dispersion equation can be set up. From it, in far-reaching analogy to Ewald's dynamical theory of X-ray reflections, the more precise position of the reflection maxima and the "resolving power" of the crystal (§ 5) as well as the reflected intensity (§ 6) can be determined. The approximation is then carried one step further (§ 7).

On the basis of the developed dynamical theory of electron diffraction, the characteristic deviations from X-ray reflection are discussed (§ 8). In connection with the very small penetrating power compared with X-rays of the same wavelength, electron diffraction deviates from X-ray diffraction in a characteristic way. The extinction laws of X-ray reflections, too, can be violated under certain circumstances. § 9 treats the case, realized by Davisson and Germer, of reflection in several directions symmetric with respect to the incident beam. The mean crystal potential acting on the electrons, responsible for the refractive index, is found on purely theoretical grounds, on the basis of wave-mechanical assumptions about the structure of the crystal atoms, to be nearly of the same magnitude as follows from the experiments (§ 10). The relation to the work function in the Richardson effect is explained. Then the Fourier coefficients of the potential variation (§ 11) are calculated, and finally with their help, in an example, the numerical values of the reflecting and resolving power, etc., are computed (§ 12). The experimental results are thereby in satisfactory agreement.

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